
Historical Information for Home Credit Ranking: A Fixed-Pipeline Development Comparison

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Abstract

Credit-risk ranking depends on both available information and the modelling pipeline used to represent it. We examined the incremental value of joined credit histories beyond application-table information in the Home Credit dataset. Logistic regression, random forest and LightGBM were evaluated on identical development folds under fixed, pipeline-specific rules. Basic historical aggregation and a further 39 behavioural features increased AUC and average precision in every recorded paired fold. Total mean-fold AUC gains were 0.021006, 0.014248 and 0.023029, respectively. A separate bounded search selected refined LightGBM and rank fusion, increasing mean AUC from 0.780230 to 0.782349. Additional feature sets failed the adoption rule, and raw fusion reduced AUC. Model diagnostics complement these comparisons but do not identify causal effects. The findings concern retrospective development comparisons: configuration selection used development data, the holdout had already been viewed, and validation-batch rank fusion differs from the empirical-distribution transformation used outside cross-validation.

1 Introduction

Payment-difficulty ranking uses information available at application time. Application attributes describe the applicant and current request; previous loans, balances and repayments may supply complementary evidence. Home Credit provides both sources [1]. A final score alone cannot distinguish gains from historical information from gains due to the modelling procedure.

We compared logistic regression (LR) with Elastic-Net regularisation [2], random forest (RF) [3] and LightGBM (LGB) [4]. They offer signed coefficients, randomised-tree ensembles and sequential boosting, respectively. This motivates an interpretability–prediction comparison, not an assumed hierarchy: our LR includes nonlinear basis terms, and model names alone do not define complexity or explanatory value.

The main question was how ranking changes from application-table information (A), to basic history (B), to behavioural enhancement (C) across three complete pipelines. Fixed original configurations and common folds evaluate each pipeline's information expansion. A separate secondary question concerns further gains from bounded LightGBM optimisation and fusion. This separation prevents tuning gains from being attributed to added history; neither a new algorithm nor independent deployment validation is claimed.

2 Data, EDA and Feature Construction

2.1 Applicants, outcomes and missingness

The labelled data contain 307,511 applications, including 24,825 positive labels (8.07%). Each row represents an applicant identified by `SK_ID_CURR`; `TARGET=1` denotes payment difficulty under the dataset’s outcome definition. IDs were used for joining and alignment, never as predictors. The official test set has 48,744 unlabelled applications, so its AUC cannot be computed locally.

Class imbalance motivated AP alongside AUC; application-field missingness motivated pipeline-specific handling. Descriptive EDA used the labelled cohort. Missing history never justified deleting applicants or replacing excluded historical inputs with fabricated zeros. Aggregation retained every applicant; learned preprocessing remained within training portions.

2.2 Three source-defined information sets

Table 1 lists shared-table column counts, including ID and label, before model-specific encoding.

Table 1: **Feature sets**

Group	Information source	Columns
A	Application-table only, with four shared derivatives	126
B	Basic history: A plus 27 historical aggregates	153
C	Enhanced history: B plus 39 behavioural features	192

A retained the original application fields, including external scores and credit-inquiry counts already present in that table. It therefore means *application-table information*, not the complete absence of credit-history-related information. Shared deterministic derivatives were age and employment duration in years, income-to-credit ratio and annuity-to-income ratio. The special employment code 365243 was handled as missing/special status, not as an ordinary duration or a direct unemployment indicator.

B aggregates bureau, previous-application, instalment, point-of-sale and credit-card histories. C adds 39 repayment, recent-history, structural and monthly-bureau features. For example, B records mean instalment-payment amount; C adds the fraction of loan-period events not fully paid by their due date, summing split payments. A recent version restricts due events to the preceding 365 days. No joined-history field, dependent ratio or history-derived missingness flag entered A. Full feature definitions remain in the notebook.

Before aggregation, the data audit excluded 17 bureau records updated after application, without changing raw CSVs or removing applicants. This specific check does not resolve every temporal ambiguity. Record counts need not count distinct loans. The legacy `INSTALL_AVG_DAYS_LATE` field means actual-payment-day offset, not delay from the scheduled date.

3 Modelling Pipelines and Evaluation

3.1 Fixed comparison and training boundaries

An ID-based, stratified split with seed 5054 assigned 292,135 applicants to development and 15,376 to a 5% holdout. All nine pipeline/information combinations used the same labels and three development fold assignments for cross-validation (CV). In each outer fold, the remaining two folds supplied training data. Imputation statistics, category vocabularies, scaling and learned basis transformations were fitted within the appropriate training portion, then applied to that fold’s validation data.

A was added retrospectively after B/C results were known, requiring nine outer and three internal early-stopping fits. The 18 historical B/C folds were reused after input, source, configuration, split and prediction checks. Saved A outer models were replay-checked; B/C outer models were not persisted, so their reuse is prediction verification, not model replay. Isolated reproduction checks did not change the reported experiments or reopen selection. The previously viewed holdout remains supplementary.

Figure 1 separates deterministic business aggregation and derivatives, which may precede splitting, from transformations learned only on training portions. Every information set uses the common ID split; outer validation receives fitted transformations without refitting. The holdout is outside this CV procedure.

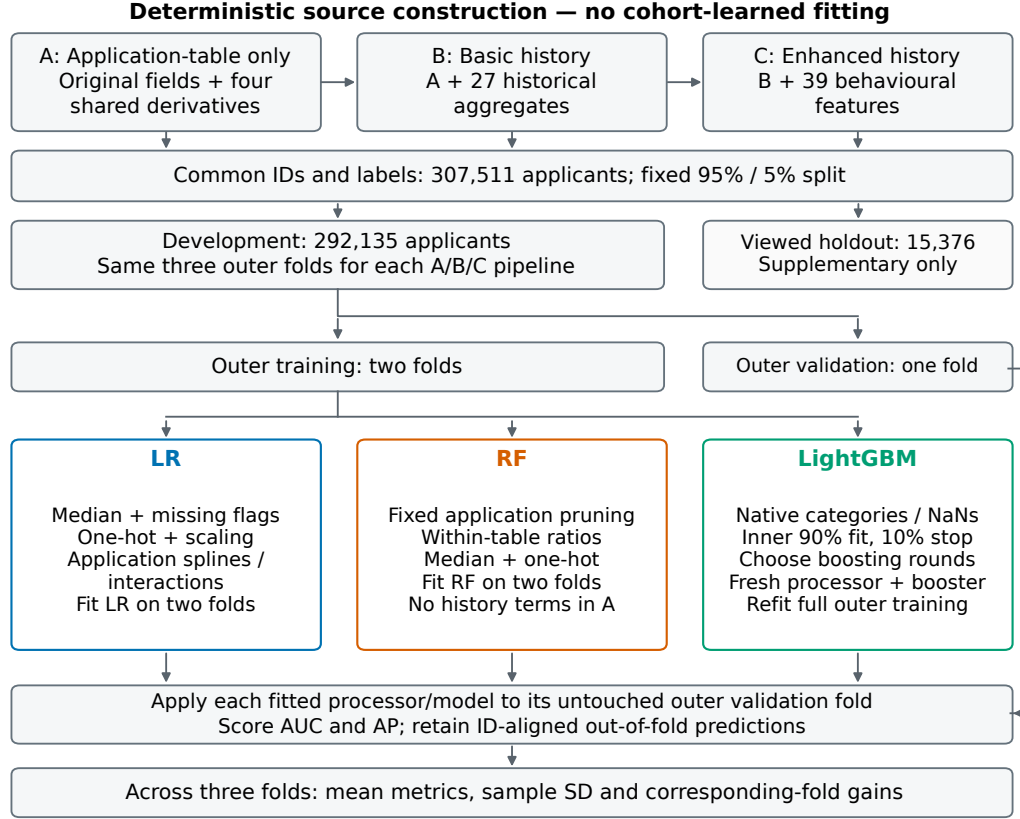


Figure 1: **Study design**

Logistic regression. LR used training-only median imputation, numeric missingness indicators, six predefined log transformations, full one-hot encoding and nonbinary scaling, with application-based splines and interactions. Fixed Elastic-Net settings were $C = 0.01$, L1 ratio 0.25 and balanced class weights. The numerical solver used equivalent coefficient splitting with bound-constrained optimisation and first-order optimality checks. Coefficients describe transformed terms, not necessarily an original variable's overall effect.

Random forest. RF used application-field pruning, within-application ratios, median imputation and one-hot encoding. History-dependent ratios and flags were excluded for A. Settings were 300 trees, depth 15, minimum leaf size 50, feature fraction 0.1, class weights 1:2, Gini criterion and seed 42. More columns increase the absolute number sampled per split, so information-set contrasts include induced representation changes.

LightGBM. The original configuration used native categorical and missing-value handling, depth 2, learning rate 0.1, at most 31 leaves, minimum leaf size 20 and balanced sample weights. A stratified internal 10% of each outer-training set, using seed 42, selected rounds by unweighted AUC with a 1,000-round cap and patience 50. The categorical processor and booster were then fitted afresh on the complete outer-training set for the chosen number of rounds. Outer validation labels did not select stopping rounds. The later refined LightGBM configuration was excluded from A/B/C. Information-value (IV) screening was disabled in the formal unified experiments.

3.2 Metrics, pairing and secondary selection

The area under the receiver operating characteristic curve (ROC AUC) measures positive–negative ranking; AP summarises precision across recall levels. Both used the natural validation distribution without class weighting. The primary summary was the unweighted mean of three fold AUCs with sample SD (denominator 3 – 1); AP followed the same convention. OOF predictions score observations excluded from their model’s training, but pooled OOF AUC was not substituted for mean fold AUC. Paired contrasts subtract corresponding folds before averaging. Overlapping training sets make fold SD descriptive, not a confidence interval; no significance test is claimed.

Secondary development selection on C covered 12 LightGBM configurations (eight structural/weight choices, four refinements), two extra feature groups and their combination over a frozen configuration range, and three prespecified fusions. The groups contain four recent–overall level differences and six Active/Closed-status delinquency summaries; overlapping windows make the former differences, not time slopes. Adoption required mean AUC gain > 0.0003 and no negative fold. Within 0.0003 of the eligible maximum, fewer components were preferred, then raw over rank fusion. These are development-selection scores, not unbiased estimates of the selected procedure.

4 Results

4.1 Gains from historical information

Historical information improved ranking in all three fixed pipelines (Figure 2). Basic history increased mean AUC by 0.009667 for LR, 0.011061 for RF and 0.014890 for LightGBM; the 39-feature stage added 0.011339, 0.003187 and 0.008138. Both stages increased AUC and AP on every recorded fold. Total C–A gains were 0.021006, 0.014248 and 0.023029. The first expansion benefited LightGBM most, whereas the second added most to LR. These contrasts support the incremental predictive value of joined history within each pipeline; they do not isolate any individual feature’s causal contribution. Table 2 reports all nine absolute AUC/AP summaries; Figure 2 shows the corresponding information gains. Full-precision fold scores remain in the notebook.

Table 2 gives three-fold means $\pm$ sample SD, rounded to four decimals. A/B/C follow Table 1; SD is not a confidence interval.

Table 2: **Development performance**

Pipeline	Metric	A: Application-only	B: Basic history	C: Enhanced history
LR	AUC	0.7529 $\pm$ 0.0025	0.7626 $\pm$ 0.0025	0.7739 $\pm$ 0.0027
	AP	0.2355 $\pm$ 0.0057	0.2472 $\pm$ 0.0045	0.2599 $\pm$ 0.0037
RF	AUC	0.7518 $\pm$ 0.0016	0.7628 $\pm$ 0.0010	0.7660 $\pm$ 0.0012
	AP	0.2340 $\pm$ 0.0050	0.2476 $\pm$ 0.0054	0.2534 $\pm$ 0.0044
LightGBM	AUC	0.7572 $\pm$ 0.0029	0.7721 $\pm$ 0.0028	0.7802 $\pm$ 0.0025
	AP	0.2420 $\pm$ 0.0051	0.2618 $\pm$ 0.0036	0.2722 $\pm$ 0.0021

Figure 2a plots AUC means with SD on a zoomed axis; b/c retain all 18 paired gains. Colours identify models; circles, squares and triangles mark folds 0, 1 and 2. Mean segments and zero-inclusive paired axes expose fold consistency.

LightGBM had the highest mean AUC in all three groups. LR ranked slightly above RF in A, slightly below it in B, and above it in C. The B difference was only about 0.000206. This changing order does not support a general rule that the more complex-looking model must predict better. These are complete-pipeline comparisons with different representations and previous search effort, not equal-budget algorithm optima.

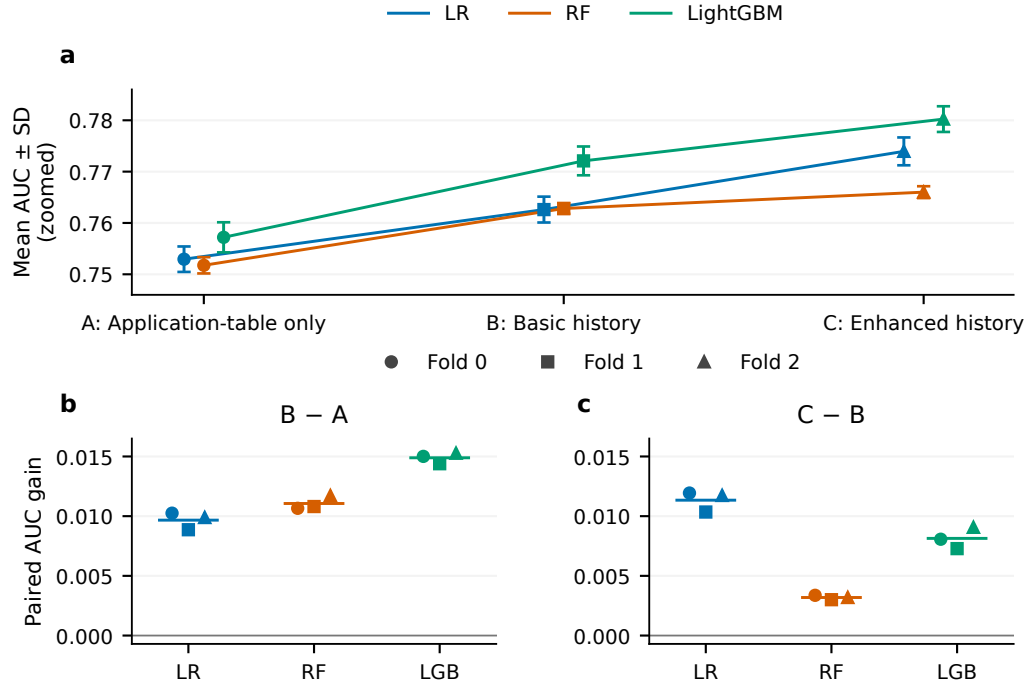


Figure 2: **Historical information gains**

4.2 Interpreting the fitted models

Figure 3 shows the five largest absolute coefficients or importances within each original C model fitted on full development data. The panels use separate scales: signed LR coefficients on transformed terms, RF impurity importance and LightGBM split gain in thousands. The tree models assign high importance to external scores already present in A; RF also uses an active net-debt-to-credit ratio. This is compatible with the A/B/C gains: importance describes a fitted model's use of correlated inputs, whereas the paired comparison evaluates an information-set expansion.

LR supplies signed coefficients on transformed terms, but mixed scaling, full dummy coding and nonlinear terms complicate original-variable interpretations. RF impurity importance and LightGBM gain identify useful splitting inputs without giving effect direction; their scales are not comparable. The LR panel includes both coefficient signs; a negative value is a conditional association for that transformed term, not evidence that changing the original variable would reduce risk. Neither coefficients nor importance values establish causality, and cross-fold importance stability was not evaluated.

4.3 Bounded optimisation

With enhanced information held fixed, the final selected base configuration increased mean AUC from 0.780230 to 0.781855 (difference 0.001625). Refined LightGBM retained depth 2, but used learning rate 0.05, minimum leaf size 100, L2 regularisation 5 and unweighted training, with a 2,000-round cap and patience 100. AP increased from 0.272194 to 0.275559. This contrast changes configuration, not the information set. Figure 4 separates this contrast from the subsequent fusion comparisons.

The parameter-only rule initially selected the feature-trial reference (AUC 0.781613; gain 0.001383). Final selection reconsidered the combined configuration/feature pool. Its maximum, the combined extra-feature candidate, raised the tolerance floor to 0.781615, excluding that reference. Refined LightGBM then won the simplicity rule without adopting extra features. Thus its selection depends on the candidate pool; the stages are not independent. The notebook reproduces both frozen choices.

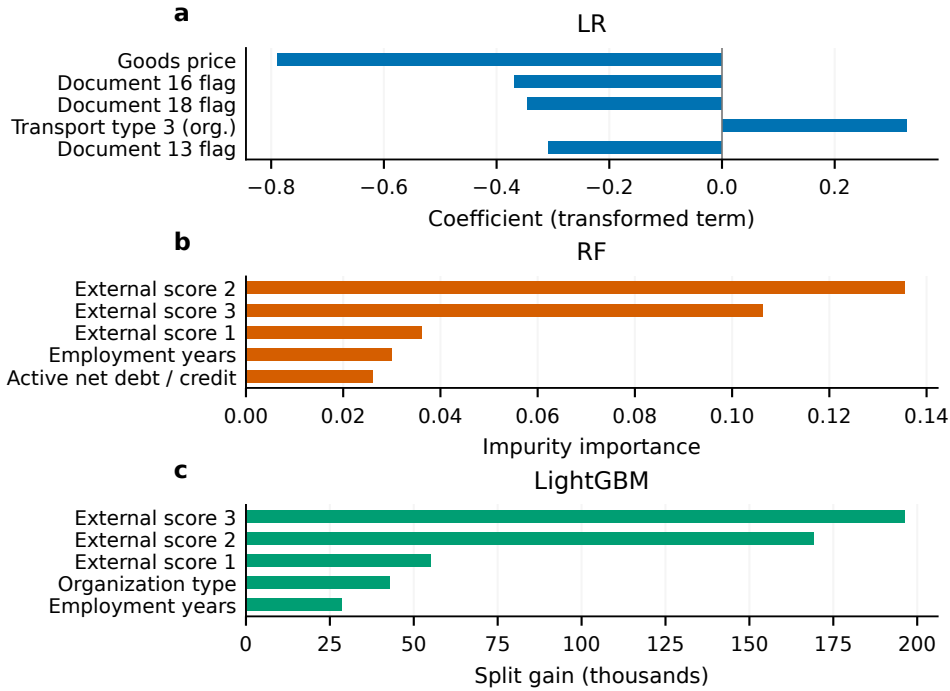


Figure 3: **Model diagnostics**

Extra-feature trials used the initial reference: depth 2, learning rate 0.1, minimum leaf size 20, L2 regularisation 5 and unweighted training. Recent-overall differences changed mean AUC by -0.000039 and status-specific summaries by -0.000109 . Their combination gained 0.000303 but declined on one fold, failing the stable-gain rule. None was adopted. Sixteen optimisation development folds were flagged as possibly truncated; final internal runs reached the 2,000-round cap. Neither a performance ceiling nor full convergence is established. Complete candidates remain in the notebook.

4.4 Fusion trade-offs

Selected LGB-LR rank fusion (80%/20%) achieved AUC 0.782349: gains of 0.000494 over refined LightGBM and 0.002119 over original C LightGBM. Raw fusion at the same weights reduced AUC on every fold despite slightly higher mean AP (Figure 4). Selected rank fusion increased mean AP only slightly and had two negative AP folds. Adoption targeted AUC, not improvement of every metric.

LGB-LR-RF rank fusion (70%/20%/10%) had higher mean AUC, 0.782638. Its 0.000289 advantage lay within the frozen 0.0003 tolerance, so the simpler two-model fusion was selected. It was the rule-selected scheme, not the highest observed score; reporting did not reopen selection.

Figure 4 shows paired AUC (a) and AP (b) changes on common development folds. The top row compares refined with original C LightGBM; the lower rows compare fusion with refined LightGBM. Weights follow model-label order. All folds, mean segments and zero are shown, without intervals. Extra-feature trials use a different reference and remain separate.

4.5 Supplementary evaluation

Table 3 reports AUC/AP on the previously viewed 15,376-applicant holdout. All five pipelines use C; the first three retain their original configurations. CDF fusion increased AUC over refined LightGBM but reduced AP. This single split supplies neither three-fold SD nor independent confirmation, and prompted no new selection.

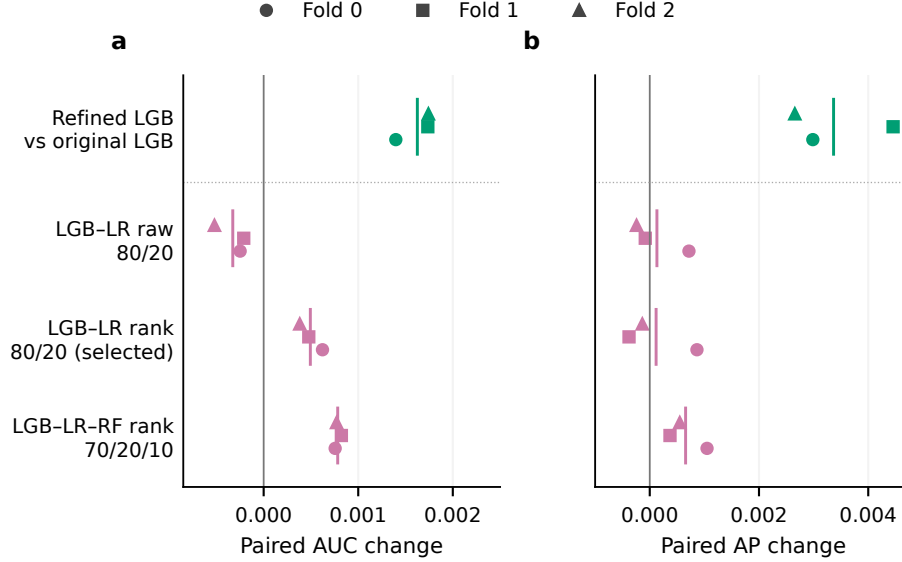


Figure 4: **Refinement and fusion**

Fusion cross-validation used percentile ranks computed inside each validation batch. Holdout and official-test scoring instead mapped model scores through development-OOB empirical cumulative distribution functions (CDFs), giving LGB-LR CDF fusion with the same 80%/20% weights. These are different scoring rules; the CV batch-ranking gains do not validate equivalent deployment-CDF performance. Fusion outputs are ranking scores, not calibrated probabilities of payment difficulty.

An author-provided Kaggle screenshot for the submission labelled “LGB-LR rank 80/20” reports a score of **0.77948** and a private score of **0.77738**, with status “Complete (after deadline)”. These supplementary scores prompted no model reselection. No official leaderboard rank is reported; the screenshot is retained with the submission materials.

Table 3: **Viewed holdout**

Pipeline	AUC	AP
LR	0.773467	0.249358
RF	0.769716	0.250608
Original LightGBM	0.778127	0.266457
Refined LightGBM	0.779527	0.269148
LGB-LR CDF fusion (80/20)	0.780285	0.266881

5 Discussion

The positive historical-information contrasts and rejected later additions show that feature value depends on the starting representation and the adoption criterion. On enhanced history, neither additional group nor their combination met the LightGBM stable-gain rule. The combined set’s positive mean gain with a negative fold illustrates why more features need not produce stable improvement. This supports selecting additions by their validation benefit rather than their number. Diminishing marginal returns are a possible interpretation for this dataset, model and limited candidate range; neither feature redundancy nor a feature-engineering ceiling was established.

The differing gains may reflect representation as well as information. Explicit repayment summaries could make useful relations more accessible to LR; RF’s smaller second-stage gain could reflect existing splits or its fixed depth and feature sampling. No component ablation distinguishes these explanations, so neither redundancy nor algorithmic capacity is established as the cause.

The LR-to-ensemble progression therefore frames an interpretability–prediction trade-off, not a universal hierarchy. LR’s mathematical structure is inspectable, yet its feature transformations restrict simple coefficient narratives. Tree ensembles expose useful-input rankings but do not make direction, individual reasoning or causality transparent. LightGBM’s stronger observed ranking does not itself resolve the practical value of these different explanatory capabilities. Calibration, fairness and lending-policy utility were not evaluated.

Several design features bound the inference. A was retrospective, development data had already supported exploration, and secondary CV selected configurations and fusion. The 5% holdout had been viewed previously. No complete nested evaluation or new external test removes this selection dependence. The three overlapping training folds provide descriptive variability, not independent experimental replication. Further, the batch-rank/CDF distinction limits transfer of the fusion result outside its evaluated scoring rule. Historical information gains exceeded later optimisation gains here, but unequal search budgets and pipeline choices prevent a general claim that feature engineering always dominates tuning.

6 Conclusion

Under fixed original pipelines and shared development folds, both basic and enhanced historical information improved payment-difficulty ranking beyond application-table information. The gains varied across pipelines, and LR/RF ordering changed with the information set. Bounded LightGBM optimisation and simple fusion added smaller observed gains, with AUC/AP trade-offs. Additional candidate features did not meet the prespecified stable-gain rule, supporting adoption based on validation benefit rather than feature count within this study. These results support a qualified development comparison of information and modelling choices, rather than statistical significance, causal risk attribution or validated deployment benefit.

Contributions and AI Use

The recorded module allocation was WU Ziyi for data preparation and EDA, JIANG Zerui for the LR baseline and team/report integration, MA Ruiyang for RF, and CHEN Kepeng for LightGBM. Final individual attribution remains subject to team confirmation. Module allocation does not imply that all implementation was individually hand-written.

AI/LLM assistance supported code inspection and repairs, documented experiments and validation, A/B/C supplementation, bounded optimisation, figures, notebook integration and report drafting. Report preparation added no candidates or selection; AI review is not human peer review or course approval. Authors remain responsible for understanding, checking, attribution and submission. The accompanying notebook documents provenance and reproduction: default Run All recomputes saved-prediction metrics with zero fits; raw reconstruction and frozen training require separate opt-in commands and original data. Executed verification levels are recorded in its README. The recorded Python environment used scikit-learn [5] and LightGBM.

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